Week4
Arrays + simulation

2026.03.17

Week4 - On-site Examination

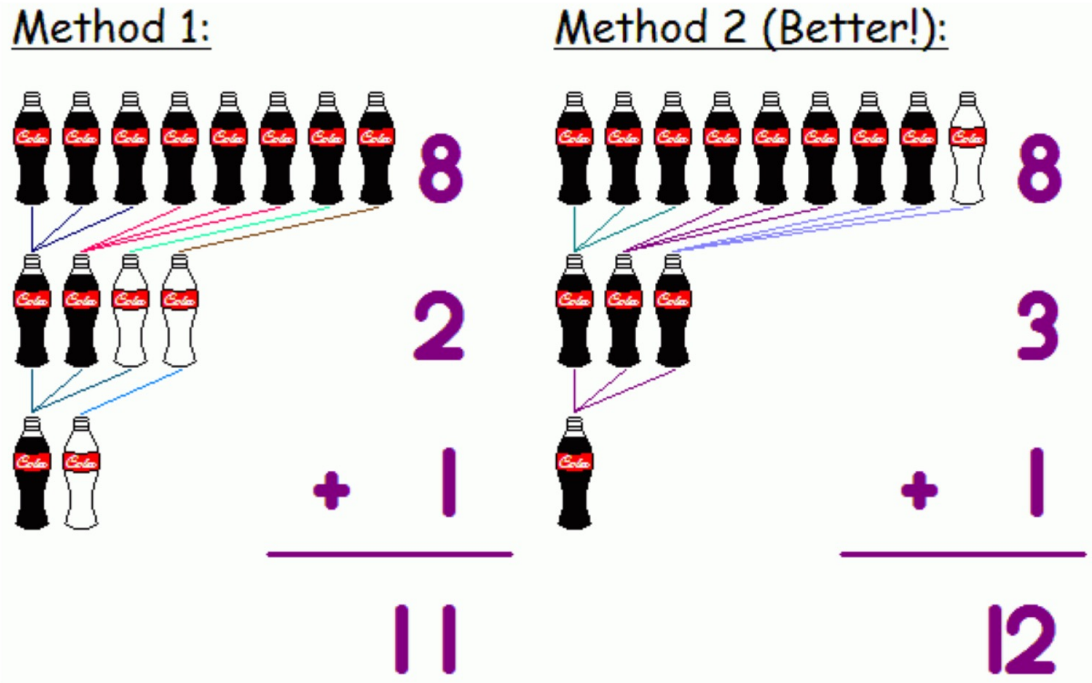
- Regular1 - UVa 11150 – Cola [UVa 11150]
- Regular2 - UVa 12503 - Robot Instructions [UVa 12503]

- Bonus1 - UVa 10908 - Large Square [UVa 10908]
- Bonus2 - UVa 10242 - Fourth Point!! [UVa 10242]

- Submission file name (e.g. 844082)
  - s{StudentID}-UVa11150.cpp → s844082-UVa11150.cpp
  - s{StudentID}-UVa12503.cpp → s844082-UVa12503.cpp
  - s{StudentID}-UVa10908-Bonus.cpp → s844082-UVa10908-Bonus.cpp
  - s{StudentID}-UVa10242-Bonus.cpp → s844082-UVa10242-Bonus.cpp
  - s{StudentID}-Week4.zip → s844082-Week4.zip (最終上傳)

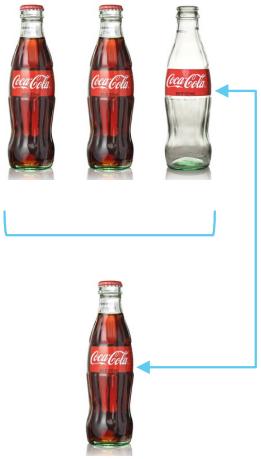
UVa 11150 - Cola

- “A bottle of Choco Cola for every 3 empty bottles returned”



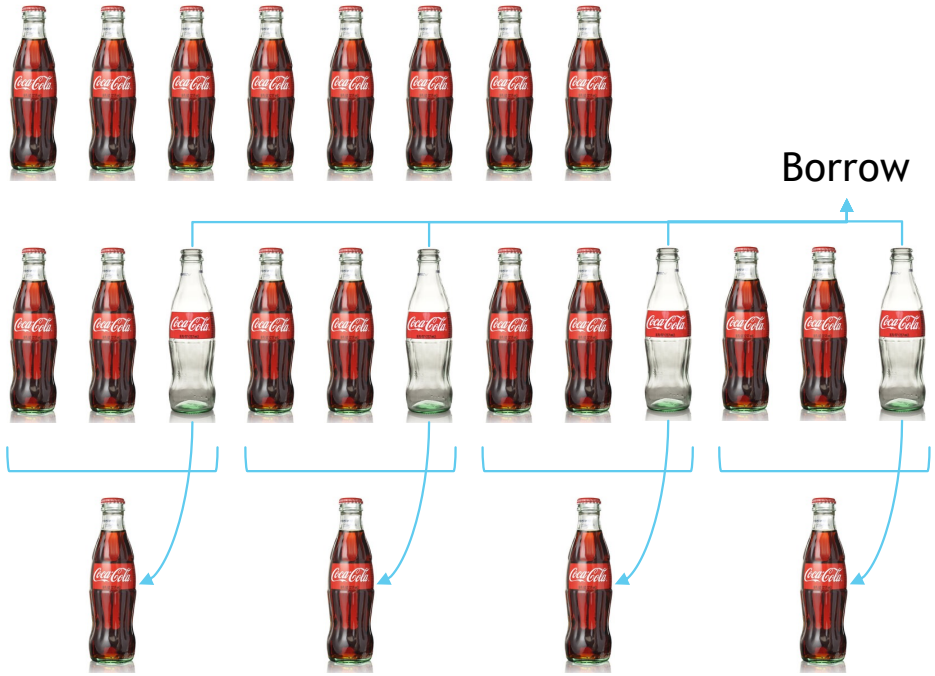
UVa 11150 - Cola

- You may **borrow empty bottles** from others, but if you do that, make sure that you have enough bottles afterwards to return to them.



borrow one empty bottles from someone (no cost)

UVa 11150 - Cola



8

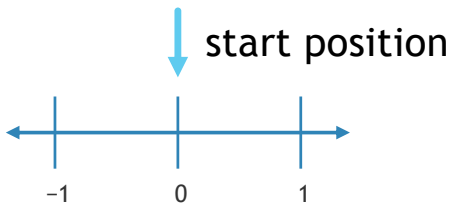
12

Leverage/開槓桿
3/2 = 1.5

UVa 12503 - Robot Instructions

- Instruction set
  - LEFT: move one unit left (decrease p by 1, where p is the position of the robot before moving)
  - RIGHT: move one unit right (increase p by 1)
  - SAME AS i: perform the same action as in the i-th instruction. It is guaranteed that i is a positive integer not greater than the number of instructions before this.

Recap: Elementary school math: Number line view

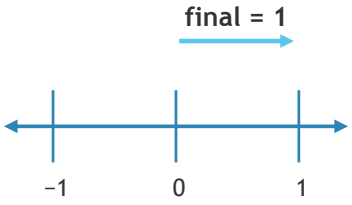


UVa 12503 - Robot Instructions

- Example

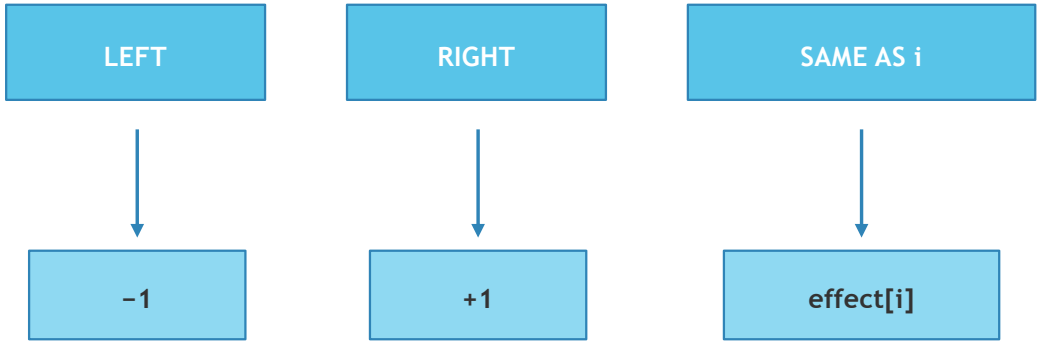
i	instruction	effect	position
1	LEFT	−1	−1
2	RIGHT	+1	0
3	SAME AS 2	+1	1

Number line view



UVa 12503 - Robot Instructions

- Reset the robot to position 0 for each test case.
- Convert each instruction into an effect: LEFT = -1, RIGHT = +1.
- For “SAME AS i”, reuse the effect of the i-th instruction.
- Store every effect in an array/vector so future instructions can refer to it.



Store the effect, then add it to the current position.

Week4 – After Class Exercise

- LeetCode: 121. Best Time to Buy and Sell Stock [LeetCode 121]
- LeetCode: 31. Next Permutation [LeetCode 31]

Week5: Before Class Exercise

- LeetCode: 88. Merge Sorted Array [LeetCode 88]
- LeetCode: 179. Largest Number [LeetCode 179]